P.2 Library of Functions; Mathematical Modeling

Name _____ Period ____ Date _____

AP Calculus P.2 · Student sheet

I can…

- identify the library functions and their properties.
- find the zeros of a polynomial, their multiplicity, and the end behavior of its graph.
- find the domain and the intercepts of a rational function.

Vocabulary · fill in the missing word

Polynomial function — a function $f(x) = a_n x^n + \cdots + a_1 x + a_0$ where n is a _____ integer

Multiplicity — the number of times a _____ $(x - r)$ appears in a factored polynomial

Rational function — a quotient of two _____ functions, $R(x) = \frac{p(x)}{q(x)}$, where q is not the zero polynomial

Floor function — $f(x) = \lfloor x \rfloor$, the _____ integer less than or equal to x

Part 1 · Crosses or touches?

Graph each one. At each zero, does the graph cross or touch the x-axis?

Function	At x = 1	At x = −2
$(x - 1)(x + 2)^2$		
$(x - 1)^2(x + 2)$		
$(x - 1)^3(x + 2)$		

Part 2 · Notes and examples (fill in each note, then work the example)

Library of functions: identity, square, cube, square root, cube root, reciprocal, and absolute value. The cube function is odd and _____ on $(-\infty, \infty)$.

Example 1

Name each function and give one property.

a. $y = x^3$

b. $y = \sqrt{x}$

c. $y = \frac{1}{x}$

d. $y = |x|$

Floor function: $\lfloor x \rfloor$ is the greatest integer _____ x.

Example 2

$f(x) = \lfloor 2x \rfloor$. Find $f(1.2)$, $f(1.6)$, and $f(-1.8)$.

Multiplicity: if $(x - r)^m$ is a factor, the graph _____ the x-axis at r when m is odd and touches it when m is even.

Example 3

$f(x) = 3(x - 7)(x + 4)^3$. List each real zero and its multiplicity, tell whether the graph crosses or touches the x-axis there, and find the y-intercept.

End behavior: for large $|x|$, a polynomial behaves like its _____ term.

Example 4

$f(x) = 4x(x^2 + 1)(x - 2)^3$. Find the real zeros with their multiplicity and describe the end behavior.

Rational functions: the domain is all real numbers except where the _____ is 0.

Example 5

Find the domain and the intercepts of $R(x) = \dfrac{3x^2 - x}{x^2 + 4}$.

Part 3 · Partner practice

Partner 1

$f(x) = -2x^2(x + 1)(x - 3)^2$. List the zeros with their multiplicity, crosses or touches, and the end behavior.

Partner 2

Find the domain and the intercepts of $H(x) = -\dfrac{4x^2}{(x - 2)(x + 4)}$.